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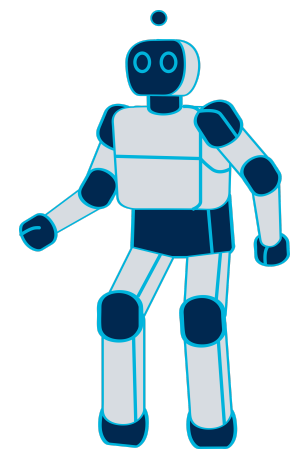
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Universal Artificial Intelligence

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Problem: Can we define, at least in theory, the "most intelligent" action to take?

In some way, the "most intelligent" program one may think of would perform **optimal decision making**. Some scientists, such as Marcus Hutter, built on Solomonoff's ideas about learning and induction to define what they call "universal artificial intelligence". Such an AI would take the best possible decisions at any time. Really?



Universal Artificial Intelligence

CREPITS

UNDERSTANDING ARTIFICIAL INTELLIGENCE
THROUGH ALGORITHMIC INFORMATION
THEORY
"UNIVERSAL ARTIFICIAL INTELLIGENCE"

Content and presentation: *Pierre-Alexandre Murena*
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NOT necessarily!

Recent studies have opened up new research directions. For instance the Universal Knowledge Seeking Agents, which don't rely on external feedback from the environment. But... they are still incomputable!

Conclusion: Universal Artificial Intelligence designates agents which can correctly build a model of the world

Hutter's AIXI instantiates an ideal version of reinforcement learning, based on algorithmic information. AIXI performs actions that maximize expected future rewards. At any moment, the expected reward results from an inductive process based on past observations and past actions. Future rewards are weighted by their probability, which depends on the complexity of the programs that generate them. Hutter's equation for "Ultimate intelligence" reads:

AIXI $a_k := \arg \max_{a_k} \sum_{o_k r_k} \dots \max_{a_m} \sum_{o_m r_m} [r_k + \dots + r_m] \sum_{q: U(q, a_1..a_m)=o_1 r_1 \dots o_m r_m} 2^{-\ell(q)}$

As Hutter puts it:

The expression shows that AIXI tries to maximize its total future reward $r_k + \dots + r_m$. If the environment is modeled by a deterministic program q , then the future perceptions $\dots o_k r_k \dots o_m r_m = U(q, a_1..a_m)$ can be computed, where U is a universal [...] Turing machine executing q given $a_1..a_m$. Since q is unknown, AIXI has to maximize its expected reward, i.e. average $r_k + \dots + r_m$ over all possible future perceptions created by all possible environments q that are consistent with past perceptions. The simpler an environment, the higher is its a-priori contribution $2^{-\ell(q)}$, where simplicity is measured by the length ℓ of program q .

Suggested reading:

- Hutter, M. (2001). Towards a universal theory of artificial intelligence based on algorithmic probability and sequential decisions. ECML-2001 (LNAI 2167), 226-238. Berlin: Springer.

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